CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 24: Measures of Central Tendency (Mean, Median, Quartiles and Mode)

Previous year board questions, topic-wise: 11 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Mean of raw and tabulated data

1. Assertion (A): The mean of the first 9 natural numbers is 4.5. Reason (R): Mean $= \frac{\text{sum of all the observations}}{\text{total number of observations}}$. (a) A true, R false (b) A false, R true (c) both true, and R is the correct reason for A (d) both true, but R is not the correct reason for A ICSE 2025, 1 mark

2. For the data below, find (a) the mean, to the nearest whole number (b) the median.

ICSE 2025, 3 marks

x	10	11	12	13	14	15	16
f	3	2	2	6	3	5	3

3. The mean of the data below is 16. Find f. ICSE 2020, 3 marks

Marks	5	10	15	20	25
Number of students	3	7	f	9	6

2 Mean of grouped data

4. The table gives the running times of some films. Find the mean running time by the step-deviation method. ICSE 2024, 3 marks

Duration (minutes)	100–110	110–120	120–130	130–140	140–150	150–160
Number of films	5	10	17	8	6	4

5. Find the mean of the marks below by the short-cut method. ICSE 2023, 3 marks

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	3	8	14	9	4	2

6. The table shows the number of patients attending a hospital on the days of a month. Find the mean by the short-cut method, taking the assumed mean as 45. Give the answer correct to 2 decimal places. ICSE 2019, 3 marks

Number of patients	10–20	20–30	30–40	40–50	50–60	60–70
Number of days	5	2	7	9	2	5

7. The mean of the data below is 24. Find a. ICSE 2018, 3 marks

Marks	0–10	10–20	20–30	30–40	40–50
Number of students	7	a	8	10	5

8. Find the mean of the data below by the step-deviation method. ICSE 2017, 4 marks

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	10	9	25	30	16	10

3 Median of raw data

9. The numbers $27, 31, 46, 52, x, x+4, 71, 79, 85, 90$ are in ascending order and their median is 64. Find x. (a) 60 (b) 61 (c) 62 (d) 66 ICSE 2024, 1 mark

10. The marks (out of 10) of 40 students in a class test are given below. Find (a) the median (b) the mode. ICSE 2019, 4 marks

Marks	1	2	3	4	5	6	7	8	9	10
Number of students	1	2	3	3	6	10	5	4	3	3

11. The marks of 10 students, in ascending order, are $13, 35, 43, 46, x, x+4, 55, 61, 71, 80$. The median is 48. Find x, and then the mode. ICSE 2017, 3 marks

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